

“Body Oracle” Translator

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Figure 1: “Body Oracle” Translator is a handheld mixed-reality mask that translates human body postures into speculative hieroglyphs, called “Body Oracle”, in real time. These AI-generated hieroglyphic characters are inspired by Chinese oracle-bone inscriptions and are derived from bodily and inter-bodily spatial positioning.

ABSTRACT

What if your body posture and movement could be read as a sentence? “Body Oracle” Translator is a handheld mixed reality device that translates human body postures and movement into speculative AI-generated hieroglyphic characters in real time. This computational linguistic speculative design project creates an alternative language system inspired by Chinese Oracle-Bone Inscriptions, promoting cross-cultural collective bodily awareness in an era dominated by language-based communication. Glimpsing through the lens, users inhabit an “alternative present”—one where these hieroglyphic characters have existed for thousands of years—where this alternative language reshapes our understanding of the body and influences collective body awareness.

Index Terms: Speculative Design, Chinese Oracle-bone Inscriptions, Somatic Awareness, Cultural Heritage, Alternative Present, Mixed Reality, AI Generation, Somaesthetics, Future Archaeology

PROJECT URL

<https://reality.design/project/body-oracle>

1 DESCRIPTION OF ARTWORK

1.1 Thematic Statement

Across both alphabetic (Western) and logographic (Eastern) traditions, most languages lack a precise lexicon for describing complex bodily postures and the body’s spatial movements. Expressing a particular pose typically demands lengthy, inefficient phrasing and remains ambiguous—especially in describing intercorporeal relations and spatial positioning. More broadly, languages lack precise vocabulary for proprioception and social proprioception, our awareness of individual and collective bodily positioning in space [6]. This limitation hinders our ability to articulate embodied cognition in today’s communication, which remains predominantly text-

based and—despite emojis—often “disembodied” [2], lacking the crucial dimension of bodily awareness.

This raises the question: *what if our language included characters that inherently heightened our awareness of body language, analogous to the way emojis convey emotion?*

In response, “Body Oracle” [4] is a computational linguistic speculative design that imagines an “alternative present” [1] in which a language system for complex proprioceptive experiences exists. Inspired by Chinese oracle-bone inscriptions [5]—one of the oldest forms of Chinese writing—we draw on their elegant pictographic affordances, in which simplified strokes mirror the objects they denote. “Body Oracle” encodes bodily postures and inter-body spatial relations using the aesthetic principles of these ancient inscriptions. The system generates AI-derived hieroglyphic characters representing corporeal poses and intercorporeal configurations. These cross-cultural, intuitive symbols—akin to emojis—aim to expand how we conceptualize, share, and communicate bodily experience.

1.2 Artwork

“Body Oracle” Translator is a handheld optical see-through mixed reality mask that converts human body postures into Body Oracle characters in real time. When a user points the translator at someone, it uses a fine-tuned YOLOv11 pose-estimation model¹ running on an iPhone to recognize body poses. The system matches these poses to characters in the “Body Oracle” Archive (a dictionary index pre-generated by the “Body Oracle” AI system, see Figure 2) and displays the corresponding character in the center of the view. An AI-generated audio of ancient Chinese pronunciation can be triggered with an on-screen button at the top of the iPhone screen. The mask is designed as a binocular, oracle-bone-shaped enclosure for HoloKit [3], a smartphone-based, open-source optical see-through stereoscopic headset (see Figure 3). The shell is fabricated from a ready-made oracle-bone form (e.g., a plastic mock-up of a turtle shell). An iPhone is mounted inside with the rear camera unobstructed, allowing environment sensing while preserving optical see-through. A hemp cord allows the device to be worn around the neck when not in use. The Translator immerses participants in an experiential alternative present, encouraging them to interact

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¹<https://docs.ultralytics.com/tasks/pose/>

with one another (see Figure 1) by making different poses in the exhibition.

This experience explores how a somatic language could enhance bodily understanding, foster collective body awareness, revitalize ancient heritage within an experiential alternative present, and point toward somaesthetic futures. Following Wittgenstein's insight—"The limits of my language mean the limits of my world"—we contend that introducing alternative somatic hieroglyphs may reshape collective semantic cognition of bodily awareness. As a sociolinguistic mediator, "Body Oracle" serves as a "true oracle" that helps us conceptualize, describe, and ultimately transform social reality through a re-embodied approach to language.

1.3 Relevance to the Theme

We present this speculative design as an experiential "future archaeology" [7] exhibition that traces the evolution of these hieroglyphs and envisions a world where proprioception forms the foundation of linguistic expression. Through this immersive "alternative present," audiences are encouraged to expand their bodily awareness. The experience creates the illusion that the Body Oracle language has existed for thousands of years, becoming deeply embedded in our culture and inhabiting a world transformed by bodily communication. This speculative "alternative present" of "future archaeology" exemplifies the IEEE VR XR Gallery's theme "(Re)Imagine: Bridging the Past, Present and Future through Immersive Experiences."

1.4 Setup for Exhibition

The installation consists of 2–3 handheld, optical see-through *Body Oracle* Translator masks (oracle-bone-shaped shells housing a HoloKit with compatible iPhone). Visitors hand-hold the mask, point it toward another person, and see the corresponding Body Oracle character overlaid at the center of view. A tap triggers an audio playback of reconstructed ancient pronunciation. Typical interaction time per visitor is 2–4 minutes, supporting both solo exploration and paired/collective play.

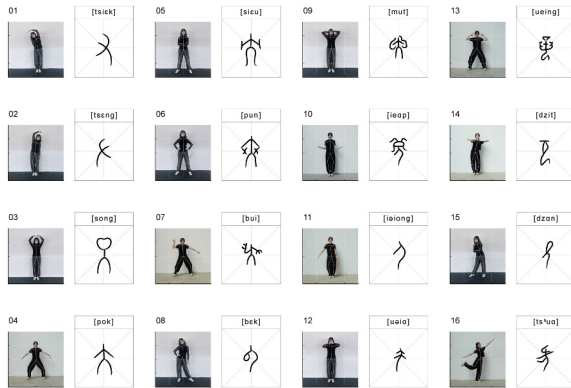


Figure 2: Dictionary Excerpts from the "Body Oracle" Archive

2 ARTIST INFORMATION

Danlin Huang is an undergraduate student at the China Academy of Art, majoring in Industrial Design. Her media artworks include "Body Oracle" and "Foldiverse." As an artist, speculative design researcher, and AR developer, her academic papers have been published in top conferences such as SIGGRAPH Asia and CHI PLAY. Her works have been exhibited at the China Digital Art Exhibition and the China National Art Exhibition, among others. Huang's



Figure 3: "Body Oracle" Translator device

work has earned her multiple accolades, including the Platinum Creative Award, the Gold Award at the Rokid University AR Innovation Challenge, the NCDA Gold Award, and the CADA Gold Award.

Ke Huang is a designer in the direction of information art design, specializing in the cross-collaborative field of designing humanities, information computing and discursive narratives. Her works have been exhibited in the 14th National Art Works Exhibition and China Digital Art Exhibition. She attended China Academy of Art and Tsinghua University for undergraduate and graduate studies respectively.

Botao 'Amber' Hu is a social computing researcher and experiential futures designer. He is currently a PhD candidate in Human Centred Computing group at University of Oxford's Department of Computer Science. He directs Reality Design Lab (<https://reality.design>), an independent interdisciplinary research and design lab. His primary focus is designing socially experiential futures using mixed reality as the main medium. He also serves as a visiting lecturer at the China Academy of Art.

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